P2 - Genetic Algorithm

Sistemas Inteligentes

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10 de marzo de 2026

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Grado en Ingeniería Informática - Especialidad Computación

1. Problem Description: MEOPT

The Mono-Objective Efficient Operation of Plug-in Hybrid Electric Trucks (MEOPT) problem requires determining the optimal driving mode assignment for a heavy-duty truck travelling a fixed route divided into N segments. The sole objective is to minimize total CO_2 emissions produced by the Internal Combustion Engine (ICE).

A solution is represented as a binary chromosome $\vec{x} \in \{0,1\}^N$, where $x_i = 1$ indicates that segment i is driven in electric mode (zero emissions, battery depleted) and $x_i = 0$ indicates driving with the ICE (CO_2 emitted, battery unaffected).

1.1. Objective Function

$$\textbf{Minimize } f_e(\vec{x}) = \sum_{i=1}^{N} e_i \cdot (1 - x_i) \tag{1}$$

where $e_i \geq 0$ [kgCO_2] is the emission cost of driving segment i with the ICE.

1.2. Energy Consumption Model

Segment costs are derived from the Vehicle Specific Power (VSP) formulation. The traction force and instantaneous power for segment i are:

$$Fd_i = \frac{\rho_{air} \cdot A_n \cdot C_{df}}{2} \cdot v_i^2 \tag{2}$$

$$Fr_i = g \cdot C_{rr} \cdot \cos(\alpha_i) \cdot m \tag{3}$$

$$Fh_i = g \cdot \sin(\alpha_i) \cdot m \tag{4}$$

$$Fa_i = a_i \cdot m \tag{5}$$

$$P_i = (Fd_i + Fr_i + Fh_i + Fa_i) \cdot v_i \tag{6}$$

Electric energy consumed when $x_i = 1$ [kWh]:

$$E_i^E = \text{máx}\left(0,\ \left(\frac{P_i}{\eta_T} + \frac{P_{aux}}{\eta_b}\right) \cdot \frac{t_i}{3{,}600 \cdot 1{,}000}\right) \tag{7}$$

Variable ICE efficiency. The combustion engine efficiency $\eta_{IC,i}$ varies per segment depending on vehicle speed and engine load, reflecting the real operating behaviour of diesel engines (peak efficiency near 60 km/h and ∼60 % of rated power):

$$f_{\text{speed},i} = \text{clamp}\left(1 - \alpha_v \cdot \left(\frac{v_i - v_{\text{opt}}}{v_{\text{scale}}}\right)^2,\ 0{,}3,\ 1{,}0\right) \tag{8}$$

$$P_{\text{norm},i} = \text{clamp}\left(\frac{|P_i|}{P_{\text{rated}}},\ 0{,}01,\ 1{,}5\right) \tag{9}$$

$$f_{\text{load},i} = \text{clamp}\left(1 - \alpha_l \cdot \left(\frac{P_{\text{norm},i} - \text{load}_{\text{opt}}}{\text{load}_{\text{scale}}}\right)^2,\ 0{,}4,\ 1{,}0\right) \tag{10}$$

$$\eta_{IC,i} = \text{clamp}\left(\eta_{IC}^{\text{peak}} \cdot f_{\text{speed},i} \cdot f_{\text{load},i},\ \eta_{IC}^{\min},\ \eta_{IC}^{\text{peak}}\right) \tag{11}$$

where $\text{clamp}(x,\ \ell,\ h) = \text{máx}(\ell, \text{mín}(h, x))$.

CO_2 emissions when $x_i = 0$ [kgCO_2]:

$$e_i = \text{máx}\left(0,\ \frac{P_i}{\eta_{IC,i}} \cdot \frac{t_i}{1{,}000} \cdot k_{\text{CO}_2}\right) \tag{12}$$

Note that negative values of P_i (e.g. downhill segments) are clamped to zero: there is no regenerative braking. Because $\eta_{IC,i}$ varies per segment, different segments carry different ICE costs, creating non-trivial trade-offs for the GA to exploit.

1.3. Battery Constraint

The battery starts fully charged at $SoE_0 = SoE_{\max}$. It depletes only during electric segments and recharges to $SoE_{\max}$ at designated stops ($b_i = 1$). A solution is feasible if and only if:

$$0 \leq SoE_i \leq SoE_{\max} \quad \forall\, i \in \{1, \ldots, N\} \tag{13}$$

Equivalently, between any two consecutive recharge points, the total electric energy demand must not exceed $SoE_{\max}$:

$$\sum_{j \in \text{leg}} E_j^E \cdot x_j \ \leq\ SoE_{\max} \tag{14}$$

2. Function Signature and Deliverable

Your submission must consist of a single MATLAB file named `solve_bin_ga.m` implementing the following interface exactly:

Listing 1: Required function signature for `solve_bin_ga.m`

```
function [best_sol, best_cost, log, num_gens] = ...
    solve_bin_ga(dist, slope, speed, time_s, accel, ...
                 bus_stop, params, time_limit)
```

2.1. Inputs

All route vectors are **row vectors of length** N (one value per segment).

Cuadro 1: Input parameters to `solve_bin_ga`

Parameter	Size	Description
`dist`	$1 \times N$	Segment distances [m]
`slope`	$1 \times N$	Slope angles [rad]
`speed`	$1 \times N$	Segment speeds [m/s]
`time_s`	$1 \times N$	Segment durations [s]
`accel`	$1 \times N$	Segment a yccelerations [m/s^2]
`bus_stop`	$1 \times N$	1 = recharge point (battery refills to $SoE_{\max}$), 0 = normal
`params`	$1 \times N$	Physical constants (first 20 entries; remainder are 0)
`time_limit`	1×1	Time budget [s] (15)

The physical constants stored in `params` are listed in Table 2.

Cuadro 2: Physical constants in `params`

Index	Symbol	Value	Units	Description
1	m	27000	kg	Truck mass
2	A_n	7	m^2	Frontal area
3	g	9.8067	m/s^2	Gravitational acceleration
4	ρ_{air}	1.225	kg/m^3	Air density
5	C_{rr}	0.006	–	Rolling resistance coefficient
6	C_{df}	0.63	–	Drag coefficient
7	P_{aux}	2000	W	Auxiliary power
8	$SoE_{\max}$	90	kWh	Maximum battery capacity
9	η_T	0.8208	–	Electric drivetrain efficiency
10	η_b	0.9	–	Battery efficiency
11	η_{IC}^{peak}	0.45	–	Peak ICE efficiency
12	k_{CO_2}	0.0002685	–	CO_2 conversion factor
13	$\eta_{IC}^{\min}$	0.15	–	Minimum ICE efficiency floor
14	v_{opt}	16.667	m/s	Optimal ICE speed (60 km/h)
15	v_{scale}	14.0	m/s	Speed half-width
16	α_v	0.6	–	Speed penalty coefficient
17	P_{rated}	350000	W	Rated engine power (350 kW)
18	load_{opt}	0.6	–	Optimal load fraction
19	$\text{load}_{\text{scale}}$	0.6	–	Load half-width
20	α_l	1.5	–	Load penalty coefficient

2.2. Outputs

Cuadro 3: Output values returned by `solve_bin_ga`

Output	Size	Description
`best_sol`	$1 \times N$	Binary chromosome of the best solution found
`best_cost`	1×1	Total CO_2 emissions [$kgCO_2$] of `best_sol`
`log`	$1 \times K$	Best cost found at each generation (convergence trace)
`num_gens`	1×1	Total number of generations completed

3. Helper Functions Available on the Server

The evaluation server provides three pre-implemented helper functions. **Do not resubmit these.**

`segment_kWh(dist_i, slope_i, speed_i, time_i, accel_i, params)` Returns the electric energy consumed [kWh] for a single segment. **Without variable efficiency normalization**.

`segment_kgCO2(dist_i, slope_i, speed_i, time_i, accel_i, params)` Returns the ICE CO_2 emissions [kgCO_2] for a single segment, using the variable η_{IC} model internally.

Skeleton

Listing 2: Recommended skeleton for `solve_bin_ga.m`

```
function [best_sol, best_cost, log, num_gens] = ...
    solve_bin_ga(dist, slope, speed, time_s, accel, bus_stop, params
        , time_limit)

    n       = length(dist);
    SoE_max = params(8);


    % --- 2. Initialise population ---
    pop_size = 50;
    pop      = randi([0 1], pop_size, n);

    best_cost = Inf;
    best_sol  = zeros(1, n);
    log       = [];
    num_gens  = 0;
    t_start   = tic;

    while toc(t_start) < time_limit - 0.5
        num_gens = num_gens + 1;

        % --- 3. Evaluate population

        % --- 4. Selection

        % --- 5. Crossover

        % --- 6. Mutation

        % --- 7. Elitism:

        % --- 8. Update best ---
        log(end+1) = best_cost;
    end
end
```

4. Deliverables

Deliverable — System Requirements

You must submit the following items through the course platform before the deadline indicated by your lecturer. The PDF must have the format **SURNAME1SURNAME2_COMPLETENAME**:

1. **Technical report** (PDF, maximum 3 pages, 12pt, a4, documentclass article):
   - **Methodology**: Describe the specific design choices made for each operaor: initialization strategy, crossover, mutation, selection and replacement operators. Pre-optimization strategy.
   - **Experimental Setup**: Specify which test instances were used for tuning and validation, and how parameter search was conducted.
   - **Results**: Present objective values and convergence behavior for the provided test cases. Include convergence plots (best cost vs. iteration) for at least the hardest instance of each problem. Discuss the effect of key parameter variations.
   - **Conclusions**: Discuss the strengths and limitations of your implementation.

   The report can be written in Spanish/English and be self-contained. Figures and tables are mandatory; do not exceed the 3-page limit.